

PROPERTY 7: LAPLACE OF INTEGRALS

$$L[f(t)] = \phi(s) \text{ then } L\left[\int_0^t f(u) du\right] = \frac{1}{s} \phi(s)$$

Example: $L\left[\frac{\sin t}{t}\right] = \cot^{-1}(s)$

Soln: $L\left[\int_0^t \frac{\sin u}{u} du\right] = \frac{1}{s} \cot^{-1}(s)$

Example: $L\left[\int_0^t u \cosh u du\right]$

Soln: $L[\cosh t] = \frac{s}{s^2 - 1}$

$$L[t \cosh t] = (-1) \frac{d}{ds} \left[\frac{s}{s^2 - 1} \right]$$

$$= (-1) \frac{(s^2 - 1)(s') - (s)(s^2 - 1)'}{(s^2 - 1)^2}$$

$$= (-1) \frac{(s^2 - 1)(1) - (s)(2s)}{(s^2 - 1)^2}$$

$$= \frac{-s^2 + 1}{(s^2 - 1)^2}$$

$$= \frac{1 - s^2}{(s^2 - 1)^2}$$

$$L\left[\int_0^t u \cosh u du\right] = \frac{1}{s} \left[\frac{1 - s^2}{(s^2 - 1)^2} \right]$$

Example: Find $\mathcal{L}\left[\int_0^t \left(\frac{1-e^{au}}{u}\right) du\right]$

Soln: $\mathcal{L}[1 - e^{at}] = \frac{1}{s} - \frac{1}{s-a}$

$$\begin{aligned}\mathcal{L}\left[\frac{1-e^{at}}{t}\right] &= \int_s^\infty \left(\frac{1}{s} - \frac{1}{s-a}\right) ds \\ &= [\log s - \log(s-a)]_s^\infty \\ &= \log\left(\frac{s}{s-a}\right) \Big|_s^\infty = \log\left(\frac{1}{1-a/s}\right) \Big|_s^\infty \\ &= \log(1) - \log\left(\frac{1}{1-a/s}\right) \\ &= 0 - \log\left(\frac{s}{s-a}\right) = \log\left(\frac{s-a}{s}\right)\end{aligned}$$

$$\mathcal{L}\left[\int_0^t \left(\frac{1-e^{au}}{u}\right) du\right] = \frac{1}{s} \log\left(\frac{s-a}{s}\right)$$

Example: Find $\mathcal{L}\left[e^{-4t} \int_0^t u \sin 3u du\right]$

Soln: $\mathcal{L}[t \sin 3t] = \frac{3(-2s)}{(s^2+9)^2} = \frac{6s}{(s^2+9)^2}$

$$\mathcal{L}\left[\int_0^t u \sin 3u du\right] = \frac{1}{s} \cdot \frac{6s}{(s^2+9)^2} = \frac{6}{(s^2+9)^2}$$

$$\mathcal{L}\left[e^{-4t} \int_0^t u \sin 3u du\right] = \frac{6}{((s+4)^2+9)^2}$$

Example: Evaluate $\int_0^{\infty} e^{-t} \left[t \int_0^t e^{-4u} \cos u \, du \right] dt$

Soln: $L \left[\left(t \int_0^t e^{-4u} \cos u \, du \right) dt \right]$

$$L[\cos t] = \frac{s}{s^2 + 1}$$

$$L[e^{-4t} \cos t] = \frac{s+4}{(s+4)^2 + 1^2}$$

$$L \left[\int_0^t e^{-4t} \cos t \, dt \right] = \frac{1}{s} \left[\frac{(s+4)}{(s+4)^2 + 1} \right]$$

$$= \frac{1}{s} \left[\frac{s+4}{s^2 + 8 + 16s + 1} \right]$$

$$= \frac{1}{s} \left[\frac{s+4}{s^2 + 6s + 9} \right]$$

$$\Rightarrow \frac{s+4}{s^3 + 8s^2 + 17s}$$

$$L \left[t \int_0^t e^{-4u} \cos u \, du \right] = (-1) \frac{d}{ds} \left[\frac{s+4}{s^3 + 8s^2 + 17s} \right]$$

$$\Rightarrow - \left[\frac{(s^3 + 8s^2 + 17s)(1) - (s+4)(3s^2 + 16s + 17)}{(s^3 + 8s^2 + 17s)^2} \right]$$

$$\Rightarrow \frac{-[12s^3 - 20s^2 - 64s - 68]}{(s^3 + 8s^2 + 17s)^2}$$

$$\Rightarrow \frac{2s^3 + 20s^2 + 64s + 68}{(s^3 + 8s^2 + 17s)^2}$$

$$\int_0^{\infty} e^{-t} \left(t \int_0^t e^{-4u} \cos u \, du \right) dt = \frac{154}{676}$$

Inverse Laplace Transform

If $L[f(t)] = \phi(s)$ then $L^{-1}[\phi(s)] = f(t)$

Standard Formulas

$$1. L[1] = \frac{1}{s} \quad L^{-1}\left[\frac{1}{s}\right] = 1$$

$$2. L[t^n] = \frac{n!}{s^{n+1}} \quad L^{-1}\left[\frac{1}{s^{n+1}}\right] = \frac{t^n}{n!}$$

$$3. L[e^{at}] = \frac{1}{s-a} \quad L^{-1}\left[\frac{1}{s-a}\right] = e^{at}$$

$$4. L[e^{-at}] = \frac{1}{s+a} \quad L^{-1}\left[\frac{1}{s+a}\right] = e^{-at}$$

$$5. L[\sin at] = \frac{a}{s^2+a^2} \quad L^{-1}\left[\frac{1}{s^2+a^2}\right] = \frac{\sin at}{a}$$

$$6. L[\cos at] = \frac{s}{s^2+a^2} \quad L^{-1}\left[\frac{s}{s^2+a^2}\right] = \cos at$$

$$7. L[\sinh at] = \frac{a}{s^2-a^2} \quad L^{-1}\left[\frac{1}{s^2-a^2}\right] = \frac{\sinh at}{a}$$

$$8. L[\cosh at] = \frac{s}{s^2-a^2} \quad L^{-1}\left[\frac{s}{s^2-a^2}\right] = \cosh at$$

Example: $L^{-1}\left[\frac{1}{s^2} - \frac{s+3}{s^2+9} + \frac{1}{s-3}\right]$

Soln: $L^{-1}\left[\frac{1}{s^2}\right] = L^{-1}[s^{-2}] = \frac{t^1}{1!}$

$$L^{-1}\left[\frac{s+3}{s^2+9}\right] = L^{-1}\left[\frac{s}{s^2+9}\right] + L^{-1}\left[\frac{3}{s^2+9}\right] = \cos 3t - \sin 3t$$

$$L^{-1}\left[\frac{1}{s-3}\right] = e^{3t}$$

Example: $L^{-1} \left[\frac{3+2s+s^2}{s^3} \right]$

Soln: $L^{-1} \left[\frac{3}{s^3} \right] + L^{-1} \left[\frac{2}{s^2} \right] + L^{-1} \left[\frac{1}{s} \right]$

$$\frac{3t^2 + 2t + 1}{13}$$

Example: $L^{-1} \left[\frac{4s+15}{16s^2-25} \right]$

Soln: $L^{-1} \left[\frac{4s}{16s^2-25} \right] + L^{-1} \left[\frac{15}{16s^2-25} \right]$

$$= \frac{4}{16} L^{-1} \left[\frac{s}{s^2 - (5/4)^2} \right] + \frac{15}{16} L^{-1} \left[\frac{1}{s^2 - (5/4)^2} \right]$$

$$\Rightarrow \frac{1}{4} \cosh \left(\frac{5}{4} t \right) + \frac{15}{16} \frac{\sinh \left(\frac{5}{4} t \right)}{\left(\frac{5}{4} \right)}$$

$$\Rightarrow \frac{1}{4} \cosh \left(\frac{5t}{4} \right) + \sinh \left(\frac{5t}{4} \right) \left(\frac{4}{5} \right)$$

Example: $L^{-1} \left[\frac{(s^2-1)^2}{s^5} \right]$

Soln: $L^{-1} \left[\frac{s^4 - 2s + 1}{s^5} \right] = L^{-1} \left[\frac{1}{s} \right] + L^{-1} \left[\frac{2}{s^4} \right] + L^{-1} \left[\frac{1}{s^5} \right]$

$$= L^{-1} \left[\frac{1}{s} \right] + 2 L^{-1} \left[\frac{1}{s^4} \right] + L^{-1} \left[\frac{1}{s^5} \right]$$

$$= 1 - 2t^3 + t^4$$

$$= \frac{1}{4} t^4 + \frac{1}{5} t^5$$

Property : Use of Shifting Theorem

$$\text{If } \mathcal{L}[f(t)] = \phi(s)$$

$$\mathcal{L}[e^{at} f(t)] = \phi(s-a)$$

$$\mathcal{L}[e^{-at} f(t)] = \phi(s+a)$$

$$\text{If } \mathcal{L}^{-1}[\phi(s)] = f(t)$$

$$\mathcal{L}^{-1}[\phi(s+a)] = e^{-at} f(t)$$

$$\mathcal{L}^{-1}[\phi(s-a)] = e^{at} f(t)$$

Example: $\mathcal{L}^{-1}\left[\frac{1}{s^2+1}\right] = \sin t$

$$\mathcal{L}^{-1}\left[\frac{1}{(s+2)^2+1}\right] = e^{-2t} \sin t = e^{-2t} \mathcal{L}^{-1}\left[\frac{1}{s^2+1}\right]$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s-3)^2+1}\right] = e^{3t} \sin t$$

$$\mathcal{L}^{-1}\left[\frac{1}{s^2}\right] = t$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s+2)^2}\right] = e^{-2t} \mathcal{L}^{-1}\left[\frac{1}{s^2}\right] = e^{-2t} t$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s-5)^2}\right] = e^{5t} \mathcal{L}^{-1}\left[\frac{1}{s^2}\right] = t e^{5t}$$

$$\mathcal{L}^{-1}\left[\frac{s}{s^2+4}\right] = \cos 2t$$

$$\mathcal{L}^{-1}\left[\frac{s}{(s-1)^2+4}\right] = \mathcal{L}^{-1}\left[\frac{(s-1)+1}{(s-1)^2+4}\right]$$

$$= \mathcal{L}^{-1}\left[\frac{s-1}{(s-1)^2+4}\right] + \mathcal{L}^{-1}\left[\frac{1}{(s-1)^2+4}\right]$$

$$= e^t \mathcal{L}^{-1}\left[\frac{s}{s^2+4}\right] + e^t \mathcal{L}^{-1}\left[\frac{1}{s^2+4}\right]$$

$$= e^t \cos 2t + e^t \sin 2t$$

Example: Find $L^{-1} \left[\frac{2s+3}{s^2+2s+2} \right]$

Soln: $L^{-1} \left[\frac{2(s+1)+1}{(s^2+1)^2+1} \right]$

$$2L^{-1} \left[\frac{s+1}{(s+1)^2+1} \right] + L^{-1} \left[\frac{1}{(s+1)^2+1} \right]$$

$$2e^{-t} L^{-1} \left[\frac{s}{s^2+1} \right] + e^{-t} L^{-1} \left[\frac{1}{s^2+1} \right]$$

$$2e^{-t} \cos t + e^{-t} \sin t$$

Example: $L^{-1} \left[\frac{3s-7}{s^2-6s+8} \right]$

Soln: $L^{-1} \left[\frac{3s-7}{(s-3)^2-1} \right]$

$$L^{-1} \left[\frac{3(s-3)+2}{(s-3)^2-1} \right]$$

$$3L^{-1} \left[\frac{s-3}{(s-3)^2-1} \right] + 2L^{-1} \left[\frac{1}{(s-3)^2-1} \right]$$

$$3e^{3t} L^{-1} \left[\frac{s}{s^2-1} \right] + 2e^{3t} L^{-1} \left[\frac{1}{s^2-1} \right]$$

$$\Rightarrow 3e^{3t} \cosh t + 2e^{3t} \sinh t$$

Example: $L^{-1} \left[\frac{s+2}{s^2-2s+17} \right]$

Soln: $L^{-1} \left[\frac{s+2}{(s-1)^2+16} \right] \Rightarrow L^{-1} \left[\frac{(s-1)+3}{(s-1)^2+16} \right]$

$$L^{-1} \left[\frac{(s-1)}{(s-1)^2+16} \right] + 3L^{-1} \left[\frac{1}{(s-1)^2+16} \right]$$

$$e^t L^{-1} \left[\frac{s}{s^2+16} \right] + 3e^t L^{-1} \left[\frac{1}{s^2+16} \right]$$

$$e^t \cos 4t + 3e^t \sin 4t$$

Example: $L^{-1} \left[\frac{s}{(s-2)^6} \right]$

Soln: $L^{-1} \left[\frac{(s-2) + 2}{(s-2)^6} \right]$

$$L^{-1} \left[\frac{s-2}{(s-2)^6} \right] + L^{-1} \left[\frac{2}{(s-2)^6} \right]$$

$$e^{2t} L^{-1} \left[\frac{1}{s^5} \right] + 2e^{2t} L^{-1} \left[\frac{1}{s^6} \right]$$

$$e^{2t} L^{-1} \left[\frac{1}{s^5} \right] + 2e^{2t} L^{-1} \left[\frac{1}{s^6} \right]$$

$$e^{2t} \frac{t^4}{15} + 2e^{2t} \frac{t^5}{16}$$

$$e^{2t} \frac{t^4 + 2t^5}{24}$$

Partial Fraction Decomposition:

$$\frac{s}{(s-2)^6} = \frac{A}{s-2} + \frac{B}{(s-2)^2} + \frac{C}{(s-2)^3} + \frac{D}{(s-2)^4} + \frac{E}{(s-2)^5} + \frac{F}{(s-2)^6}$$

$$s = A(s-2)^5 + B(s-2)^4 + C(s-2)^3 + D(s-2)^2 + E(s-2) + F$$

$$s = A(s^5 - 10s^4 + 40s^3 - 80s^2 + 80s - 32) + B(s^4 - 8s^3 + 24s^2 - 32s + 16) + C(s^3 - 6s^2 + 12s - 8) + D(s^2 - 4s + 4) + E(s - 2) + F$$

$$s = As^5 - 10As^4 + 40As^3 - 80As^2 + 80As - 32A + Bs^4 - 8Bs^3 + 24Bs^2 - 32Bs + 16B + Cs^3 - 6Cs^2 + 12Cs - 8C + Ds^2 - 4Ds + 4D + Es - 2E + F$$

$$s = As^5 + (-10A+B)s^4 + (40A-8B+C)s^3 + (-80A+24B-6C+D)s^2 + (80A-32B+12C-4D+E)s + (-32A+16B-8C+4D-2E+F)$$

$$\begin{cases} A = 0 \\ -10A+B = 0 \Rightarrow B = 0 \\ 40A-8B+C = 0 \Rightarrow C = 0 \\ -80A+24B-6C+D = 0 \Rightarrow D = 0 \\ 80A-32B+12C-4D+E = 1 \Rightarrow E = \frac{1}{4} \\ -32A+16B-8C+4D-2E+F = 0 \Rightarrow F = \frac{1}{2} \end{cases}$$

Final Answer:

$$L^{-1} \left[\frac{s}{(s-2)^6} \right] = e^{2t} \left(\frac{t^4}{15} + \frac{2t^5}{16} \right) = e^{2t} \frac{t^4 + 2t^5}{24}$$

PROPERTY : METHOD OF PARTIAL FRACTION

$$\frac{1}{(s+3)(s+2)} = \frac{A}{(s+3)} + \frac{B}{(s+2)}$$

$$\frac{s}{(s^2+4)(s^2+9)} = \frac{As+B}{s^2+4} + \frac{Cs+D}{s^2+9}$$

$$\frac{s}{(s^2+16)(s+3)} = \frac{As+B}{s^2+16} + \frac{C}{s+3}$$

$$\frac{1}{(s^2+9)(s+2)^3} = \frac{As+B}{s^2+9} + \frac{C}{s+2} + \frac{D}{(s+2)^2} + \frac{E}{(s+2)^3}$$

Example: $\frac{3s+1}{(s+1)(s^2+2)}$

Soln: $\frac{3s+1}{(s+1)(s^2+2)} = \frac{A}{(s+1)} + \frac{Bs+C}{s^2+2}$

$$3s+1 = A(s^2+2) + Bs+C(s+1)$$

$$3s+1 = As^2 + 2A + Bs^2 + Bs + Cs + C$$

$$3s+1 = (A+B)s^2 + (B+C)s + (2A+C)$$

$$A+B=0$$

$$B+C=3$$

$$2A+C=1$$

$$A = -2/3 \quad B = 2/3 \quad C = 7/3$$

$$\frac{3s+1}{(s+1)(s^2+2)} = \frac{-2}{3} \times \frac{1}{s+1} + \frac{2}{3} \times \frac{s}{s^2+2} + \frac{7}{3} \times \frac{1}{s^2+2}$$

$$\mathcal{L}^{-1} \left[\frac{3s+1}{(s+1)(s^2+2)} \right] = \frac{-2}{3} \mathcal{L}^{-1} \left[\frac{1}{s+1} \right] + \frac{2}{3} \mathcal{L}^{-1} \left[\frac{s}{s^2+2} \right] + \frac{7}{3} \mathcal{L}^{-1} \left[\frac{1}{s^2+2} \right]$$

$$= \frac{-2}{3} e^{-t} + \frac{2}{3} \cos \sqrt{2}t + \frac{7}{3} \frac{\sin \sqrt{2}t}{\sqrt{2}}$$

Example: $\frac{s+2}{(s+3)(s+1)^2} = \frac{A}{s+3} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$

Soln: $s+2 = A(s+1)^2 + B(s+3)(s+1) + C(s+3)(s+1)$

Put $s = -1$ $1 = 2D \Rightarrow D = 1/2$

Put $s = -3$ $-1 = 8A \Rightarrow A = -1/8$

Put $s = 0$ $2 = A + 3B + 3C + 3D$

$\frac{2}{8} + \frac{1}{8} + 3B + 3C + 3D = 11/8$

$\frac{2}{8} + \frac{1}{8} + 3B + 3C + 3D = 11/8$

$3B + 3C = 3 \Rightarrow B + C = 1$

$s = 1, 3 = 8A + 16B + 8C + 4D$

$3 = 1 + 16B + 8C + 2 \Rightarrow 16B + 8C = 0$

$2B + C = 0 \Rightarrow B = -1/8, C = 1/4$

$\Rightarrow \frac{1}{8} \frac{1}{(s+3)} - \frac{1}{8} \frac{1}{(s+1)} + \frac{1}{4} \frac{1}{(s+1)^2} + \frac{1}{2} \frac{1}{(s+1)^3}$

$\Rightarrow \frac{1}{8} e^{-3t} - \frac{1}{8} e^{-t} + \frac{1}{4} t e^{-t} + \frac{1}{2} t^2 e^{-t}$

Example: $\frac{s}{(s^2+1)(s^2+4)(s^2+9)}$

Soln: Put $s^2 = x$ $1 = A(x+4)(x+9) + B(x+1)(x+9)$

Put $x = -1$ $1 = 24A \Rightarrow A = 1/24$

Put $x = -4$ $1 = 15B \Rightarrow B = -1/15$

Put $x = -9$ $1 = 40C \Rightarrow C = 1/40$

$\frac{1}{(s^2+1)(s^2+4)(s^2+9)} = \frac{1}{24} \left(\frac{1}{s^2+1} \right) - \frac{1}{15} \left(\frac{1}{s^2+4} \right) + \frac{1}{40} \left(\frac{1}{s^2+9} \right)$

$$\frac{s}{(s^2+1)(s^2+4)(s^2+9)} = \frac{1}{24} \left(\frac{s}{s^2+1} \right) - \frac{1}{15} \left(\frac{s}{s^2+4} \right) + \frac{1}{40} \left(\frac{s}{s^2+9} \right)$$

$$= \frac{1}{24} \cos t - \frac{1}{15} \cos 2t - \frac{1}{40} \cos 3t$$

Example: $\frac{5s^2 - 15s - 11}{(s+1)(s-2)^2} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$

Soln: $\frac{5s^2 - 15s - 11}{(s+1)(s-2)^2} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$

$$5s^2 - 15s - 11 = A(s-2)^2 + B(s+1)(s-2) + C(s+1)$$

Put $s = -1 \Rightarrow 9 = 9A \Rightarrow A = 1$

Put $s = 2 \Rightarrow -21 = 4C \Rightarrow C = -\frac{21}{4}$

Put $s = 0 \Rightarrow -11 = 4A - 2B + C$

$$-11 = 4(1) - 2B + (-\frac{21}{4}) \Rightarrow -11 = 4 - 2B - \frac{21}{4} \Rightarrow -11 = \frac{16 - 21}{4} - 2B \Rightarrow -11 = -\frac{5}{4} - 2B \Rightarrow -2B = -11 + \frac{5}{4} = -\frac{39}{4} \Rightarrow B = \frac{39}{8}$$

$$\frac{5s^2 - 15s - 11}{(s+1)(s-2)^2} = \frac{1}{s+1} + \frac{39}{8(s-2)} - \frac{21}{4(s-2)^2}$$

$$\mathcal{L}^{-1} \left[\frac{5s^2 - 15s - 11}{(s+1)(s-2)^2} \right] = e^{-t} + \frac{39}{8} e^{2t} - \frac{21}{4} t e^{2t}$$

Example: $s^2 + 2s + 4$

$$(s^2 + 2s + 5)(s^2 + 2s + 2)$$

Soln: let $s^2 + 2s = x$

$$x - 4 = A(x + 2) + B(x + 5)$$

$$\text{Put } x = -2$$

$$-6 = A(0) + B(-2 + 5)$$

$$-6 = 3B$$

$$-2 = B$$

$$\text{Put } x = -5$$

$$-9 = A(-3) + 0$$

$$3 = A$$

$$x - 4$$

$$= \frac{3}{x+2} - \frac{2}{x+5}$$

$$(x+5)(x+2)$$

$$(x+5)$$

$$(x+2)$$

$$s^2 + 2s - 4$$

$$(s^2 + 2s + 5)(s^2 + 2s + 2)$$

$$= 3e^{-t} \left[-i \left[\frac{1}{s^2 + u} \right] - 2e^{-t} \left[-i \left[\frac{1}{s^2 + 1} \right] \right]$$

$$= \frac{3}{a} e^{-t} \sin 2t - 2 e^{-t} \sin t$$

Example: s^2

$$(s^2 + 1)(s^2 + 4)$$

Soln: put $s^2 = x$

$$x = B(x + 1) + A(x + 4)$$

$$-1 = 3A$$

$$A = -1/3$$

$$B = 4/3$$

$$\text{Put } x = -4$$

$$x = B(x + 1) + A(x + 4)$$

$$-4 = -3B$$

Putting the values

$$x = \frac{4}{3}(x + 1) + \left(-\frac{1}{3}\right)(x + 4)$$

$$\frac{s^2}{(s^2 + 1)(s^2 + 4)} = \frac{4}{3} \left(\frac{1}{x + 1} \right) + \left(-\frac{1}{3}\right) \left(\frac{1}{x + 4} \right)$$

$$= \frac{1}{3} \sin t + \frac{4}{3} \frac{\sin t}{2}$$

Convolution Theorem

$$\mathcal{L}^{-1}[\phi_1(s)\phi_2(s)] = \int_0^t f_1(u)f_2(t-u)du$$

where $f_1(u) = \mathcal{L}^{-1}[\phi_1(s)]$

$f_2(u) = \mathcal{L}^{-1}[\phi_2(s)]$

Corollary: $\mathcal{L}^{-1}\left[\frac{1}{s}\phi(s)\right] = \int_0^t f(u)du$

where $f(t) = \mathcal{L}^{-1}[\phi(s)]$

Example: $\mathcal{L}^{-1}\left[\frac{s^2}{(s^2+4)^2}\right]$

Soln: $\frac{s^2}{(s^2+4)^2} = \left(\frac{s}{s^2+4}\right)\left(\frac{s}{s^2+4}\right)$

$\phi_1(s) \cdot \phi_2(s)$

$f_1(t) = \mathcal{L}^{-1}[\phi_1(s)] = \cos 2t$

$f_2(t) = \mathcal{L}^{-1}[\phi_2(s)] = \cos 2t$

$$\mathcal{L}^{-1}\left[\frac{s^2}{(s^2+4)^2}\right] = \int_0^t f_1(u)f_2(t-u)du$$

$$= \int_0^t \cos 2u \cos 2(t-u) du$$

$$= \frac{1}{2} \int_0^t [\cos 2t + \cos(4u-2t)] du$$

$$= \frac{1}{2} \left[\cos 2t(u) \Big|_0^t + \frac{\sin(4u-2t)}{4} \Big|_0^t \right]$$

$$= \frac{1}{2} \left[t \cdot \cos 2t + \frac{1}{4} [\sin 2t - \sin(-2t)] \right]$$

$$= \frac{1}{2} \left[t \cdot \cos 2t + \frac{1}{4} \cdot \sin 2t \right]$$

Example:

$$(s-1)(s^2+4)$$

Soln: $\phi_1(s) = \frac{1}{s^2+4}$

$\phi_2(s) = \frac{1}{s-1}$

$$S_1 = L^{-1}[\phi_1(s)] = \frac{\sin 2t}{2}$$

$$S_2 = L^{-1}[\phi_2(s)] = e^t$$

$$L^{-1}\left[\frac{1}{(s-1)(s^2+4)}\right] = \int_0^t f_1(u) f_2(t-u) du$$

$$= \int_0^t \frac{\sin 2u}{2} \cdot e^{(t-u)} du$$

$$= \frac{e^t}{2} \int_0^t e^{-u} \sin 2u du$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2} [a \sin bx - b \cos bx]$$

$$= \frac{e^t}{2} \left[\frac{e^{-u}}{(-1)^2 + 2^2} (-\sin 2u - 2 \cos 2u) \right]_0^t$$

$$= \frac{e^t}{2} \left[\frac{e^{-t}}{5} (-\sin 2t - 2 \cos 2t) - \frac{1}{5} (-\sin 0 - 2 \cos 0) \right]$$

$$= \frac{-1}{10} [\sin 2t + 2 \cos 2t] + \frac{e^t}{5}$$

Example: 1

$$s(s^2+9)$$

Soln: $\phi_1(s) = \frac{1}{s^2+9}$, $\phi_2(s) = \frac{1}{s}$

$$f_1(t) = \mathcal{L}^{-1}[\phi_1(s)] = \frac{\sin 3t}{3}$$

$$f_2(t) = \mathcal{L}^{-1}[\phi_2(s)] = 1$$

$$\mathcal{L}^{-1}\left[\frac{1}{s(s^2+9)}\right] = \int_0^t f_1(u) f_2(t-u) du$$

$$= \int_0^t \frac{\sin 3u}{3} du$$

$$= \frac{1}{3} \int_0^t \sin 3u du$$

$$= \frac{1}{3} \left[-\frac{\cos 3u}{3} \right]_0^t$$

$$= -\frac{1}{9} [-\cos 3t - 1]$$

Example: 1

$$(s-2)(s+2)^2$$

Soln: $\phi_1(s) = \frac{1}{(s+2)^2}$, $\phi_2(s) = \frac{1}{s-2}$

$$f_1(t) = \mathcal{L}^{-1}[\phi_1(s)] = t e^{-2t}$$

$$f_2(t) = \mathcal{L}^{-1}[\phi_2(s)] = e^{2t}$$

$$\left[\frac{1}{(s-2)(s+2)^2} \right] = \int_0^t f_1(u) \cdot f_2(t-u) du$$

$$= \int_0^t u \cdot e^{-2u} \cdot e^{2(t-u)} du$$

$$= e^{2t} \int_0^t u \cdot e^{-4u} du$$

$$= e^{2t} \int_0^t (u \cdot v) \leftarrow \text{Integrate by parts}$$

$$\int UV = UV_1' - U'V_2 + U''V_3$$

$$= e^{2t} \left[U \left(\frac{e^{-4t}}{-4} \right) - 1 \left(\frac{e^{-4t}}{16} \right) \right]_0^t$$

$$= e^{2t} \left[-\frac{te^{-4t}}{4} - \frac{e^{-4t}}{16} - 0 + \frac{1}{16} \right]$$

$$= \frac{e^{2t}}{16} - \left[\frac{te^{-4t}}{4} \right] - \frac{e^{-2t}}{16}$$

Example: $\frac{s^2 + s}{(s^2 + 1)(s^2 + 2s + 2)} \Rightarrow \frac{s(s+1)}{(s^2 + 1)(s^2 + 2s + 2)}$

Soln: $\phi_1(s) = \frac{s+1}{s^2 + 2s + 2}$ $\phi_2(s) = \frac{s}{s^2 + 1}$

$$f_1(t) = L^{-1}[\phi_1(s)]$$

$$= L^{-1} \left[\frac{s+1}{(s+1)^2 + 1} \right]$$

$$= e^{-t} \cdot L^{-1} \left[\frac{s}{s^2 + 1} \right] \Rightarrow e^{-t} \cdot \cos t$$

$$f_2(t) = L^{-1}[\phi_2(s)]$$

$$= L^{-1} \left[\frac{s}{s^2 + 1} \right] = \cos t$$

$$L^{-1} \left[\frac{s^2 + s}{(s^2 + 1)(s^2 + 2s + 2)} \right] = \int_0^t f_1(u) \cdot f_2(t-u) du$$

$$= \int_0^t e^{-u} \cdot \cos u \cdot \cos(t-u) du$$

$$= \int_0^t e^{-u} \left[\cos t + \frac{\cos(2u-t)}{2} \right] du$$

$$\Rightarrow \frac{1}{2} \left\{ \int_0^t e^{-u} \cos t \, du + \int_0^t e^{-u} \cos (2u - t) \, du \right\}$$

$$\Rightarrow \frac{1}{2} \left\{ \cos \left(\frac{e^{-u}}{-1} \right) \Big|_0^t + \left[\frac{e^{-u}}{(-1)^2 + 2^2} \left\{ -\cos (2u - t) + 2 \sin (2u - t) \right\} \right]_0^t \right\}$$

$$\int e^{ax} \cos bx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx - b \sin ax]$$

$$\Rightarrow \frac{1}{2} \left\{ \cos t (-e^{-t} - (-1)) + \frac{e^{-t}}{5} (-\cos t + 2 \sin t) - 1 (-\cos t - 2 \sin t) \right\}$$

Example: $\frac{s}{(s^2+1)(s^2+4)} = \frac{s \cdot 1}{(s^2+1)(s^2+4)}$

Soln: $\phi_1(s) = \frac{s}{s^2+4}$ $\phi_2(s) = \frac{1}{s^2+1}$

$$s_1 = L^{-1}[\phi_1(s)] = \cos 2t$$

$$s_2 = L^{-1}[\phi_2(s)] = \sin t$$

$$L^{-1} \left[\frac{1}{(s^2+1)(s^2+4)} \right] = \int_0^t f_1(u) f_2(t-u) \, du$$

$$= \int_0^t \cos 2u \sin (t-u) \, du$$

$$= \frac{1}{2} \int_0^t \sin (t+u) + \sin (t-3u) \, du$$

$$= \frac{1}{2} \left[-\cos (t+u) - \frac{\cos (t-3u)}{-3} \right]_0^t$$

$$= \frac{1}{2} \left[(-\cos 2t + \frac{1}{3} \cos 2t) - (-\cos t + \frac{1}{3} \cos t) \right]$$

$$= \frac{1}{2} \left[-\frac{2}{3} \cos 2t + \frac{2}{3} \cos t \right] = \frac{1}{3} [\cos t - \cos 2t]$$

Property : Use of Differentiation of $\phi(s)$

$$\mathcal{L}^{-1}[\phi(s)] = -\frac{1}{t} \mathcal{L}^{-1}[\phi'(s)]$$

Example: Find $\mathcal{L}^{-1}\left[\log\left(\frac{s-a}{s-b}\right)\right]$

Soln: $\phi(s) = \log\left(\frac{s-a}{s-b}\right) = \log(s-a) - \log(s-b)$

$$\phi'(s) = \frac{1}{s-a} - \frac{1}{s-b}$$

$$\mathcal{L}^{-1}[\phi'(s)] = e^{at} - e^{bt}$$

$$\mathcal{L}^{-1}[\phi(s)] = -\frac{1}{t} \mathcal{L}^{-1}[\phi'(s)]$$

$$= -\frac{1}{t} [e^{at} - e^{bt}]$$

$$= \frac{e^{bt} - e^{at}}{t}$$

Example: Find $\mathcal{L}^{-1}\left[\tan^{-1}\left(\frac{s}{2}\right)\right]$

Soln: $\phi(s) = \tan^{-1}\left(\frac{s}{2}\right)$

$$\phi'(s) = \frac{1}{1 + (s/2)^2} \left(\frac{1}{2}\right)$$

$$\phi'(s) = \frac{1}{s^2 + 4} \left(\frac{1}{2}\right) = \frac{1}{2(s^2 + 4)}$$

$$\mathcal{L}^{-1}[\phi'(s)] = \frac{1}{2} \mathcal{L}^{-1}\left[\frac{1}{s^2 + 4}\right] = \frac{1}{2} \times \frac{\sin 2t}{2}$$

$$= \frac{\sin 2t}{4}$$

$$\therefore L^{-1}[\phi(s)] = -\frac{1}{t} L^{-1}[\phi'(s)]$$

$$(p+s)^2 \phi - (1) = \frac{-1}{t} \sin 2t \quad \text{put } p=2 \Rightarrow \phi = \frac{-1}{(s+2)^2}$$

Example: $L^{-1}[\cot^{-1}(s+1/2)]$

Soln: $\phi(s) = \cot^{-1}\left(\frac{s+1}{2}\right)$

$$\phi'(s) = \left[\frac{-1}{1+x^2} \right] \Rightarrow \phi'(s) = \frac{-1}{1+(s+1/2)^2} \cdot \frac{1}{2}$$

$$= -\frac{4}{4+(s+1)^2} \left(\frac{1}{2} \right) = \frac{-2}{(s+1)^2+4}$$

$$L^{-1}[\phi'(s)] = -2 L^{-1}\left[\frac{1}{(s+1)^2+4} \right]$$

$$= -2 e^{-t} L^{-1}\left[\frac{1}{s^2+4} \right]$$

$$= -2 \cdot e^{-t} \cdot \frac{\sin 2t}{2}$$

$$= -e^{-t} \sin 2t$$

$$L^{-1}[\phi(s)] = -\frac{1}{t} L^{-1}[\phi'(s)]$$

$$= -\frac{1}{t} (-e^{-t} \sin 2t)$$

$$= \frac{e^{-t} \sin 2t}{t}$$

Example: $L^{-1} \left[\log \left(\frac{s^2+4}{s^2+9} \right) \right]$

Soln: $\phi(s) = \log \left(\frac{s^2+4}{s^2+9} \right) = \log(s^2+4) - \log(s^2+9)$

$$\phi'(s) = \left[\frac{s}{s^2+4} - \frac{s}{s^2+9} \right] 2$$

$$L^{-1}[\phi'(s)] = 2[\cos 2t - \cos 3t]$$

$$L^{-1}[\phi(s)] = -\frac{1}{t} [t^{-1}(s)]$$

$$= -\frac{1}{t} [2\cos 2t - 2\cos 3t]$$

$$= \frac{2\cos 3t - 2\cos 2t}{t}$$

low chance

Example: $L^{-1} [\tan^{-1} (2/s^2)]$

Soln: $\phi(s) = \tan^{-1} \left[\frac{2}{s^2} \right]$

$$\phi'(s) = \frac{1}{1 + \left(\frac{2}{s^2} \right)^2} \left(\frac{-4}{s^3} \right)$$

$$\phi'(s) = -4 \left(\frac{s}{s^4+4} \right)$$

$$L^{-1}[\phi'(s)] = -4 L^{-1} \left[\frac{s}{s^4+4} \right]$$

$$= -4 L^{-1} \left[\frac{s}{(s^4+4s^2+4) - 4s^2} \right]$$

$$= -4 L^{-1} \left[\frac{s}{(s^2+2)^2 - (2s)^2} \right]$$

$$= -4L^{-1} \left[\frac{S}{(s^2+2s+2)(s^2-2s+2)} \right]$$

By doing partial fractions,

$$= -4L^{-1} \left[\frac{1}{4} \left[\frac{1}{s^2-2s+2} - \frac{1}{s^2+2s+2} \right] \right]$$

$$= -L^{-1} \left[\frac{1}{s^2-2s+2} - \frac{1}{s^2+2s+2} \right]$$

$$= -L^{-1} \left[\frac{1}{(s-1)^2+1} - \frac{1}{(s+1)^2+1} \right]$$

$$= - \left\{ e^t L^{-1} \left[\frac{1}{s^2+1} \right] - e^{-t} L^{-1} \left[\frac{1}{s^2+1} \right] \right\}$$

$$L^{-1}[\phi(s)] = - \{ e^t \sin 2t - e^{-t} \sin t \}$$

$$L^{-1}[\phi(s)] = -\frac{1}{t} L^{-1}[\phi(s)] = -\frac{1}{t} \sin t (e^{-t} - e^t)$$

$$= 2 \sin t \cdot \sin ht$$

Solving Differential Equations using LT

$$\mathcal{L}[f'(t)] = -f(0) + s\mathcal{L}[f(t)]$$

$$\mathcal{L}[f''(t)] = -f'(0) + sf'(0) + s^2\mathcal{L}[f(t)]$$

$$\mathcal{L}[f'''(t)] = -f''(0) + sf''(0) - s^2f(0) + s^3\mathcal{L}[f(t)]$$

$$\mathcal{L}[y'] = -y(0) + s\mathcal{L}[y] = -y(0) + s\bar{y}$$

$$\mathcal{L}[y''] = -y'(0) - sy(0) + s^2\bar{y}$$

$$\mathcal{L}[y'''] = -y''(0) - sy'(0) - s^2y(0) + s^3\bar{y}$$

Example: $3\frac{dy}{dt} + 2y = e^{3t}$ $y=1$ at $t=0$

Soln: $3y' + 2y = e^{3t}$

$$3\mathcal{L}[y'] + 2\mathcal{L}[y] = \mathcal{L}[e^{3t}]$$

$$3[-y(0) + s\mathcal{L}[y]] + 2\bar{y} = \frac{1}{s-3}$$

$$3[-1 + s\bar{y}] + 2\bar{y} = \frac{1}{s-3}$$

$$(3s+2)\bar{y} - 3 = \frac{1}{s-3}$$

$$(3s+2)\bar{y} = \frac{1}{s-3} + 3 = \frac{3s-8}{s-3}$$

$$\bar{y} = \frac{3s-8}{(s-3)(3s+2)}$$